Mind the Gap

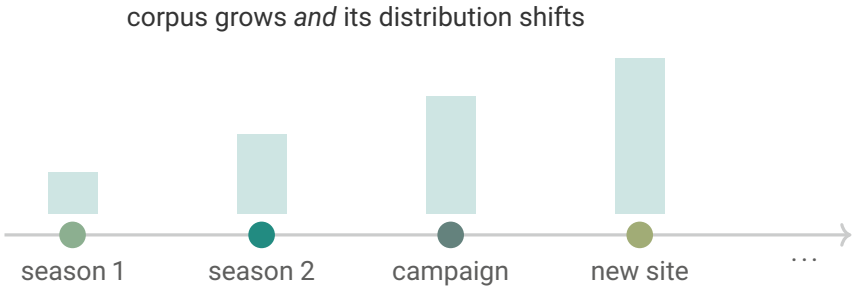
Joint vs. Data-Incremental Self-Supervised Pretraining for Plant Classification

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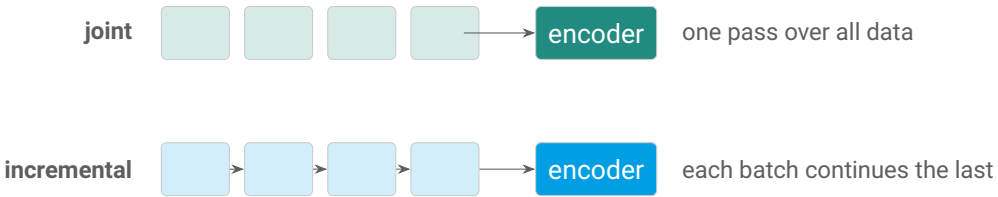
Agricultural image collections are never finished



Every arrival poses the same question: what do we do with the encoder?

Deep learning as the default tool for such imagery: Kamilaris and Prenafeta-Boldú [1]

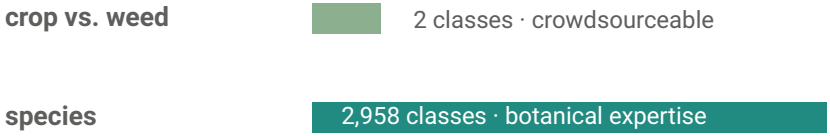
Two ways to absorb new data



Incremental is cheaper and does not presuppose that earlier images are still available.

What does it cost in representation quality?

Labels are not equally expensive

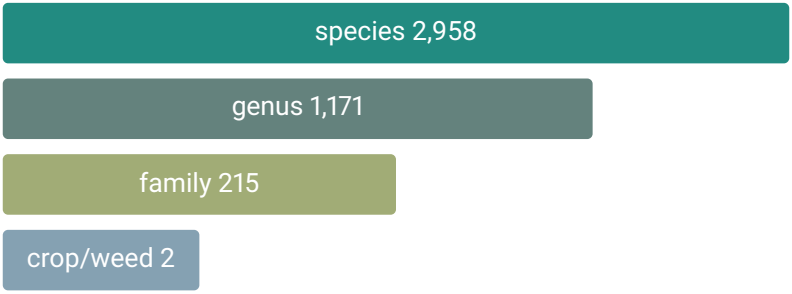


Self-supervised pretraining decouples the two: train the encoder on **unlabelled** images, freeze it, then fit a small probe for whichever target you can afford to label.

Horn et al. [2]; Jain et al. [3]; SSL on agricultural imagery: Güldenring and Nalpantidis [4]

Does the cost of pretraining *incrementally*
depend on the granularity of the labels
you later evaluate on?

One encoder, four nested targets

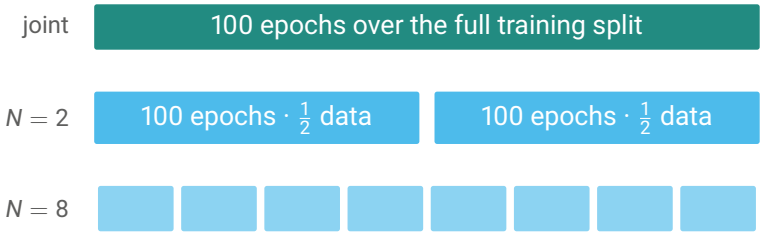


560,846	2,958	75 / 15 / 10
images	plant species	split, stratified by species

The hierarchy is strictly nested, so a single frozen encoder serves all four probes.

iNatAg-mini: Jain et al. [3]; taxonomy metadata: Joshi et al. [5]

A fair comparison: identical compute



219,000
optimizer steps, both arms

100×
every image seen, both arms

Only the *order* in which the data arrives differs. DINOv2 ViT-S/14, trained from random initialization.

DINOv2: Oquab et al. [6]; instance-incremental streams: Douillard and Lesort [7]

Two ways to cut the stream

i.i.d.

Every segment contains all 2,958 species.
Isolates the effect of *sequential presentation* alone.

100 % coverage

in every segment

seasonal

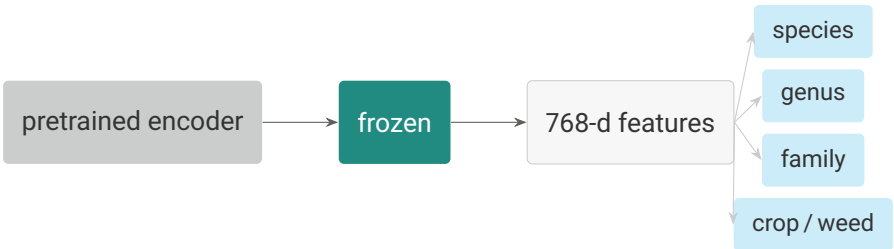
Ordered by month of capture, hemisphere corrected.
Deliberately introduces *covariate shift*.

87.4–94.0 %

species coverage at $N = 8$

Ordering by absolute date was measured and rejected: 50.9 % of the corpus falls in a single year.

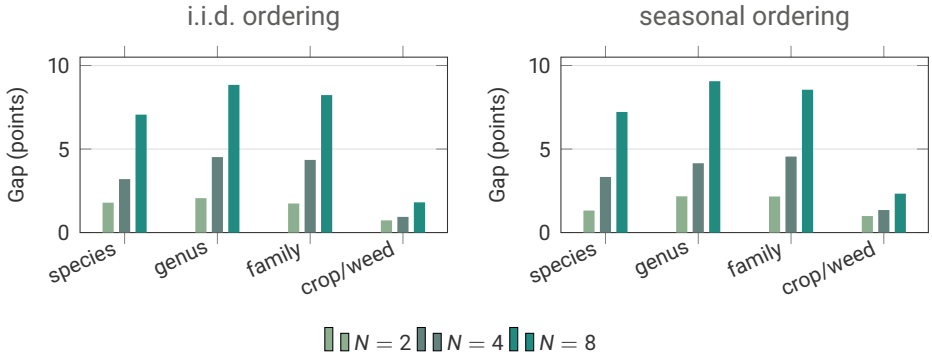
Evaluation: freeze, then probe



Linear probes only. The encoder is never fine-tuned, so every difference comes from the representation itself.

Reported as micro-averaged top-1 accuracy.

The penalty depends on granularity



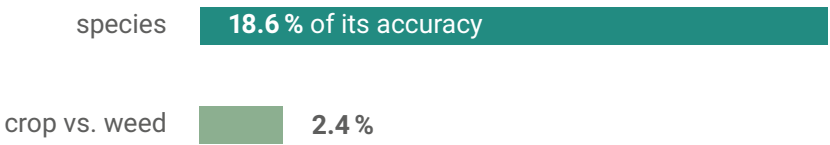
7.06 points

species, at $N = 8$

1.81 points

crop vs. weed, at $N = 8$

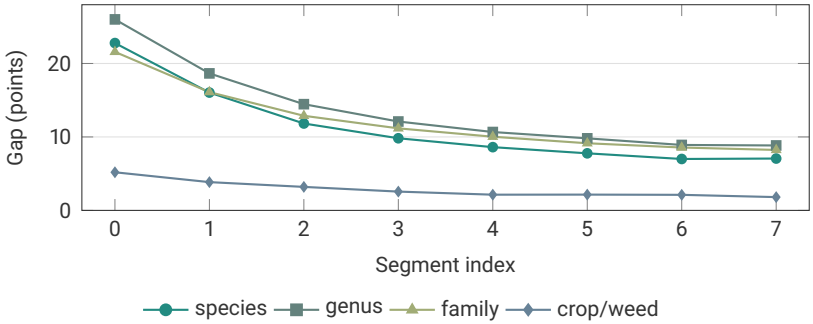
Eight times the damage, same encoder



Relative to the jointly trained baseline, at eight segments.

The gap grows monotonically with segment count, and the two orderings agree closely.

Most of the deficit is recovered, but not all



The large initial deficit reflects limited data, not forgetting. The residual is what the previous slide quantifies.

The representation drifts

$1.49 \rightarrow 1.75$
mean taxonomic distance of a species error

$19.0 \rightarrow 14.6\,\%$
errors staying inside the correct genus

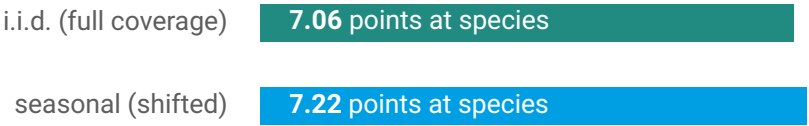
$0.667 \rightarrow 0.607$
CKA against the jointly trained encoder

All three move monotonically with segment count, and all three point the same way.

Errors do not merely become more frequent, they land further from the truth in the taxonomy.

Centred kernel alignment: Kornblith et al. [8]

Fragmentation, not distribution shift



Breaking coverage barely changes the outcome.

Within the range of shift this corpus admits, the penalty is driven by *how often the stream is cut*, not by what is in each piece.

What this means in practice

1

Coarse tasks tolerate incremental updates.

Crop vs. weed loses 1.81 points, exactly where labels are cheap and updates frequent.

2

Fine-grained targets do not.

Species forfeits close to a fifth of its accuracy, and further segments do not recover it.

3

One aggregate score hides this.

Averaging the four levels reports about six points and conceals both ends.

Where one frozen encoder serves both, the finest downstream target dictates the update strategy.

Aggregate reporting is the norm in continual SSL: Fini et al. [9]; Purushwalkam et al. [10]; Madaan et al. [11]

Limitations and what comes next

Bounds on these conclusions

Single seed per configuration; the claim rests on the monotone trend replicated under two orderings.

The schedule restarts at every boundary, so the measured gap is an *upper bound* on the ordering effect.

560k images, one architecture, one objective.

Next steps

How small a replay buffer closes the species gap, and does that budget itself depend on granularity?

Does distillation preserve fine structure, or only lift the aggregate?

Hierarchy-aware objectives, and the full 4.7M-image release.

Distillation for continual SSL: Fini et al. [9]; full release: Jain et al. [3]

References

[1] A. Kamilaris and F. X. Prenafeta-Boldú. **“Deep Learning in Agriculture: A Survey”.** In: *Computers and Electronics in Agriculture* 147 (2018), pp. 70–90.

[2] G. V. Horn et al. **“The iNaturalist Species Classification and Detection Dataset”.** In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2018, pp. 8769–8778.

[3] N. Jain et al. ***iNatAg: Multi-Class Classification Models Enabled by a Large-Scale Benchmark Dataset with 4.7M Images of 2,959 Crop and Weed Species*.** 2025.

[4] R. Güldenring and L. Nalpantidis. **“Self-Supervised Contrastive Learning on Agricultural Images”.** In: *Computers and Electronics in Agriculture* 191 (2021), p. 106510.

[5] A. Joshi, D. Guevara, and M. Earles. **“Standardizing and Centralizing Datasets for Efficient Training of Agricultural Deep Learning Models”.** In: *Plant Phenomics* 5 (2023), p. 0084.

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[6] M. Oquab et al. **“DINOv2: Learning Robust Visual Features without Supervision”.** In: *Transactions on Machine Learning Research* (2024).

[7] A. Douillard and T. Lesort. ***Continuum: Simple Management of Complex Continual Learning Scenarios*.** 2021.

[8] S. Kornblith et al. **“Similarity of Neural Network Representations Revisited”.** In: *Proceedings of the 36th International Conference on Machine Learning*. Vol. 97. Proceedings of Machine Learning Research. 2019, pp. 3519–3529.

[9] E. Fini et al. **“Self-Supervised Models are Continual Learners”.** In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 9611–9620.

[10] S. Purushwalkam, P. Morgado, and A. Gupta. **“The Challenges of Continuous Self-Supervised Learning”.** In: *Computer Vision – ECCV 2022*. Lecture Notes in Computer Science. Cham: Springer, 2022, pp. 702–721.

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[11] D. Madaan et al. **“Representational Continuity for Unsupervised Continual Learning”.** In: *International Conference on Learning Representations*. 2022.

Thank you for your attention.

Questions?

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